UPCAT REVIEWER

ANSWER KEY AND SOLUTION TO

UPCAT MATHEMATICS REVIEWER (60 ITEMS)

If you find errors, please let me know (reviewermath@yahoo.com)

ANSWER KEY

1	D	13	C	25	D	37	D	49	B
2	C	14	C	26	C	38	A	50	C
3	A	15	D	27	BONUS	39	D	51	D
4	B	16	C	28	B	40	B	52	B
5	C	17	D	29	B	41	C	53	C
6	B	18	D	30	A	42	D	54	C
7	B	19	C	31	A	43	C	55	D
8	D	20	A	32	A	44	C	56	A
9	B	21	D	33	D	45	C	57	C
10	C	22	C	34	A	46	D	58	B
11	A	23	B	35	A	47	A	59	A
12	D	24	C	36	C	48	A	60	D

SOLUTION

1. Amanda wants to buy a graphing calculator that is already on sale for P4200. The sale price is 30% below the original price. If she is able to get an additional 15% off for being a student, how much money was saved from the original price?

(A) P1,890.00 (B) P1,170.00 (C) P2,000.00 (D) P2,430.00

Solution:

100%-30%

To get the original price divide P4200 by 0.7 the result is P6,000.00.

100%-15%

Amanda paid P4200 x 0.85 which is equal to P3,570.00

Amanda saved P6,000 – P3,570.00 = P2,430.00.

Answer: D

2. Let r and s be the roots of the quadratic equation

$$(x-2)(x-3)+(x-3)(x+1)+(x+1)(x-2)=0.$$

Evaluate: $\dfrac{1}{(r+1)(s+1)}+\dfrac{1}{(r-2)(s-2)}+\dfrac{1}{(r-3)(s-3)}$.

(A) $1/4$ (B) $3/17$ (C) $8/11$ (D) $4/15$

Solution :

If we simplify the quadratic equation, the result is

$(x^2-5x+6)+(x^2-2x-3)+(x^2-x-2)=0.$

$3x^2-8x+1=0.$

$r+s=8/3$ and $rs=1/3$.

Simplify $\dfrac{1}{(r+1)(s+1)}+\dfrac{1}{(r-2)(s-2)}+\dfrac{1}{(r-3)(s-3)}$

$$\frac{1}{rs+(r+s)+1}+\frac{1}{rs-2(r+s)+4}+\frac{1}{rs-3(r+s)+9}$$

$$=\frac{1}{\frac{1}{3}+\frac{8}{3}+1}+\frac{1}{\frac{1}{3}-2\left(\frac{8}{3}\right)+4}+\frac{1}{\frac{1}{3}-3\left(\frac{8}{3}\right)+9}$$

$$=\frac{1}{\frac{1}{3}+\frac{8}{3}+1}+\frac{1}{\frac{1}{3}-2\left(\frac{8}{3}\right)+4}+\frac{1}{\frac{1}{3}-3\left(\frac{8}{3}\right)+9}=\frac{1}{4}+\frac{3}{-11}+\frac{1}{4}=\frac{8}{11}$$

Recall:

If r and s are the roots of $ax^2+bx+c=0$ then

sum of roots = $r+s=\dfrac{-b}{a}$

product of roots = $rs=\dfrac{c}{a}$

Answer : C

3. Solve for x: $25\left(30^{x}\right)=\left(6^{3}\right)\left(5^{5}\right)$

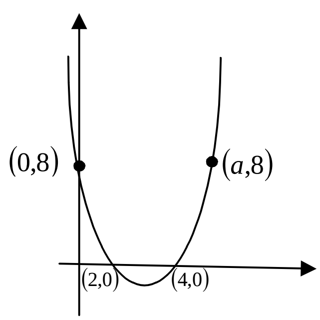
(A) 3 (B) 4 (C) 6 (D) 8

Solution:

$$25\left(30^{x}\right)=\left(6^{3}\right)\left(5^{5}\right)$$
$$\Rightarrow 5^{2}\left(5^{x}\cdot 6^{x}\right)=\left(6^{3}\right)\left(5^{5}\right)$$ divide both sides by 5^{2}
$$\Rightarrow 5^{x}\cdot 6^{x}=\left(6^{3}\right)\left(5^{3}\right)$$
$$\Rightarrow 30^{x}=30^{3}$$
$$\Rightarrow x=3$$

Answer: A

4. The x-intercepts of a parabola are 2 and 4, and the y-intercept is 8. If the parabola passes through the point (a,8), what is the value of a?



A) 5 (B) 6 (C) 7 (D) 8

Solution:

$y = k(x-2)(x-4)$, when x = 0, $y = k(0-2)(0-4) \Rightarrow 8k = 8 \Rightarrow k = 1$

The equation of the parabola is $y = (x-2)(x-4)$

$(a-2)(a-4) = 8$

$\Rightarrow a^2 - 6a + 8 = 8$

$\Rightarrow a^2 - 6a = 0$

$\Rightarrow a(a-6) = 0$

$\Rightarrow a = 0$ or $a = 6$,

a = 6

Answer: B

5. Sa isang lungsod, 2/5 ng mga lalaking may sapat na gulang ay kasal sa 5/7 ng babaeng may sapat na gulang. Ang bilang ng mga kasal na kalalakihan at kasal na kababaihan ay pantay-pantay, at ang mga may sapat na gulang ng populasyon ay higit sa 3400. Ano ang pinakamaliit na posibleng bilang ng mga may sapat na gulang na residente sa lungsod?

(A) 3442 (B) 3452 (C) 3432 (D) 3412

Solution:

Let x = number of men

y = number of women

$\frac{2}{5}x = \frac{5}{7}y$

So $x = \frac{25}{14}y$

Also $y = \frac{14}{25}x$

x and y are integers
so $y = 14n$ for a certain integer n
this implies that $x = 25n$

$$x + y > 3400$$
$$25n + 14n > 3400$$
$$39n > 3400$$
$$n > \frac{3400}{39} = 87.18$$

The smallest possible integer n is 88.

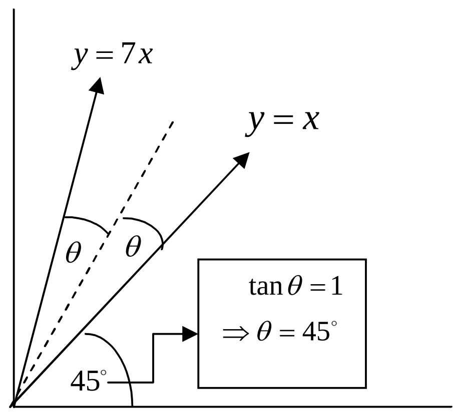
Therefore $x + y = 39n = 39 \times 88 = 3432$

Answer: C

6. An **acute angle** is formed by two lines of slope 1 and 7. What is the slope of the line which bisects this angle?

(A) $3/2$ (B) 2 (C) 3 (D) 4

Solution:



If the angle of inclination of the line $y = mx + b$ is θ then

$m = \tan\theta$

The angle of inclination of the line $y = 7x$ is $\left(2\theta + 45^{O}\right)$

therefore $\tan\left(2\theta + 45^{O}\right) = 7$

$$\frac{\tan 2\theta + \tan 45^{O}}{1 - \tan 2\theta \cdot \tan 45^{O}} = 7$$

$$\frac{\tan 2\theta + 1}{1 - \tan 2\theta} = 7$$

$$\tan 2\theta + 1 = 7 - 7\tan 2\theta$$
$$\tan 2\theta = {}^{3}\!/_{4}$$

Recall:

$$\tan(A+B) = \frac{\tan A + \tan B}{1 - \tan A \cdot \tan B}$$

half-angle tangent formulas

$$\tan\left(\frac{B}{2}\right) = \frac{1-\cos B}{\sin B} = \frac{\sin B}{1+\cos B}$$

Using half-angle tangent formula

$$\tan\theta = \tan\left(\frac{2\theta}{2}\right)$$
$$= \frac{1-\cos 2\theta}{\sin 2\theta}$$

[$\tan 2\theta = {}^{3}\!/_{4}$, using Pythagorean triple 3-4-5, $\cos 2\theta = {}^{4}\!/_{5}$ and $\sin 2\theta = {}^{3}\!/_{5}$]

$$= \frac{1 - {}^{4}\!/_{5}}{{}^{3}\!/_{5}}$$

$$= \frac{1}{3}$$

The angle of inclination of the angle bisector is $\left(\theta + 45^{O}\right)$
Therefore the slope of the angle bisector is equal to $\tan\left(\theta + 45^{O}\right)$

$$\tan\left(\theta + 45^{\circ}\right) = \frac{\tan\theta + \tan 45^{O}}{1 - \tan\theta \cdot \tan 45^{O}}$$

$$= \frac{\tan\theta + 1}{1 - \tan\theta}$$

$$= \frac{\frac{1}{3} + 1}{1 - \frac{1}{3}}$$

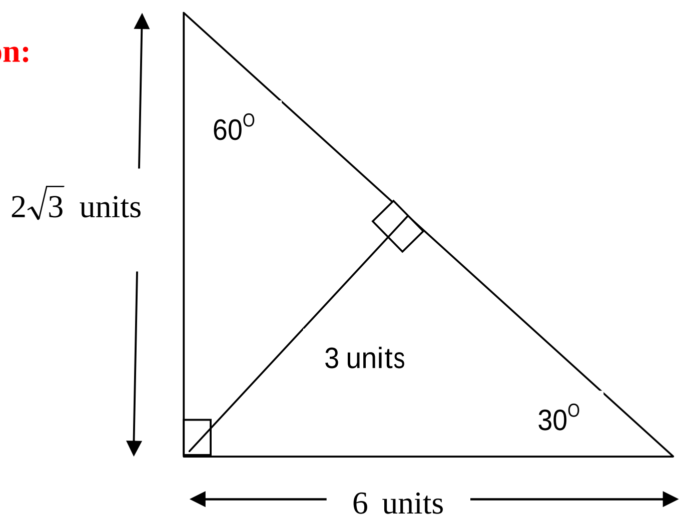
$$= 2$$

Answer: B

7. The altitude to the hypotenuse of a triangle with angles of 30° and 60° is 3 units. What is the area of the triangle in square units?

(A) $9\sqrt{2}$ (B) $6\sqrt{3}$ (C) $9\sqrt{3}$ (D) $6\sqrt{2}$

Solution:



base = 6 units

height = $2\sqrt{3}$ units

Area = $\frac{1}{2} \times b \times h$

$= \frac{1}{2} \times 6 \times 2\sqrt{3}$

$= 6\sqrt{3}$ square units

Answer: B

8. What was the Biblical approximation to π?

(A) 3.14 (B) $\frac{22}{7}$ (C) 3.1416 (D) 3

Answer: D

9. In the freshman class at Uno High School, there are 18 boys and 12 girls. The average height of the boys is 170 cm, and that of the girls is 160 cm. What is the average height of all the students in the class?

(A) 165 cm (B) 166 cm (C) 167 cm (D) 168 cm

Average height = $\frac{18 \times 170 + 12 \times 160}{18 + 12} cm$

= $166\ cm$

Answer: B

10. Having found that $(x-2)$ is a factor of $P(x)=3x^3+2x^2-x-30$, which of the following is another factor of $P(x)$?
(A) $x+3$
(B) $x-10.6$
(C) $3x^2+8x+15$
(D) none of the above

Solution:

We divide $P(x)$ by $x-2$
We use synthetic division

$$
\begin{array}{c|cccc}
 & 3 & 2 & -1 & -30 \\
2 & & 6 & 16 & 30 \\
\hline
 & 3 & 8 & 15 & 0
\end{array}
$$

$3x^2+8x+15$ is not factorable

Answer: C

11. Gawing simple: $\left(\sqrt{2}+\sqrt{6}\right)\div\sqrt{2+\sqrt{3}}$.
(A) 2 (B) 3 (C) 4 (D) 6

Solution:

$$\frac{\sqrt{2}+\sqrt{6}}{\sqrt{2+\sqrt{3}}}\bullet\frac{\sqrt{2-\sqrt{3}}}{\sqrt{2-\sqrt{3}}}$$
$$=\frac{\sqrt{\left(\sqrt{2}+\sqrt{6}\right)^2\left(2-\sqrt{3}\right)}}{\sqrt{4-3}}$$
$$=\sqrt{\left(8+4\sqrt{3}\right)\left(2-\sqrt{3}\right)}$$

$$= \sqrt{4\left(2+\sqrt{3}\right)\left(2-\sqrt{3}\right)}$$

$$= \sqrt{4\left(4-3\right)}$$

$$= 2$$

Answer: A

12. Nicole and Chris have the same grade average before the final exam. The final exam counted as 20% of the semester grade. Nicole got 7 percentage points higher than Chris on her semester grade, and the final exam was worth 100 points. What is the number of points in the positive difference between their final exam grades?

(A) 30 (B) 25 (C) 50 (D) 35

Solution:

Let x be the number of points that Nicole got on her final exam
and y be the number of points that Chris got on his final exam.

Then

$0.2\left(x-y\right)=7$ multiply both sides by 5

$x-y=35$

Answer: D

13. Suppose there is a row of lockers numbered 1 through 50 with all the doors closed. A student is going to walk up and down the lockers opening and closing the locker doors. If a door is closed the student will open it, and if it is open, it will be closed .If the student starts with multiple of one, and continues through multiples of 50. How many lockers are open when the student is finished?

(A) 5 (B) 10 (C) 7 (D) 8

Solution:

If a number is a perfect square it has an odd number of factors if not it has an even number of factors.

Example is 9 the factors of 9 are 1,3,9 there are 3 factors and 3 is an odd number
12 is not a perfect square. The factors of 12 are 1,2,3,4,6,12, there are 6 factors and 6 is an even number

Locker # 9 : multiple of 1 (open), multiple of 3 (close), multiple of 9 (open)

All lockers with a perfect square number are open when the student is finished the remaining lockers are closed.

Locker # 12: multiple of 1(open), multiple of 2 (close), multiple of 3 (open), multiple of 4 (close), multiple of 6 (open), multiple of 12 (close)

The number of perfect squares less than 50 is 7 because $7^2 < 50 < 8^2$.

7 lockers are open.

Answer: C

14. Ang isang mag-aalahas ay may 3 spheres ng ginto. Ang sukat ng kanilang radius ay 3mm, 4mm,at 5mm .Kung ang lahat ng tatlong spheres ay tinunaw at bumuo ng panibagong sphere na ginto. Ano ang sukat ng radius ng bagong sphere na ginto?

(A) 8mm (B) 9mm (C) 6 mm (D) 10 mm

Solution:

	radius	Volume = $\frac{4}{3}\pi r^3$
1^{st} sphere	$3\,mm$	36π
2^{nd} sphere	$4\,mm$	$\frac{256}{3}\pi$
3^{rd} sphere	$5\,mm$	$\frac{500}{3}\pi$
	Total Volume:	$36\pi + \frac{256}{3}\pi + \frac{500}{3}\pi = 288\pi$

To get the radius of the new sphere we solve for r in the following equation:

$$\frac{4}{3}\pi r^3 = 288\pi$$

$$\Rightarrow r^3 = 288\pi$$

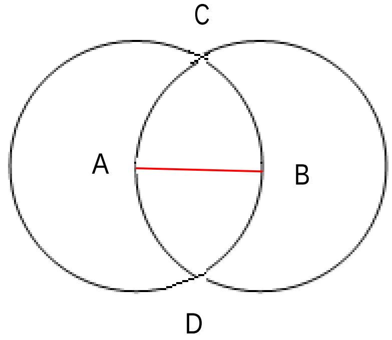
$$\Rightarrow r^3 = 216$$

$$\Rightarrow r = 6$$

The radius of the new sphere is 6 mm.

Answer: C

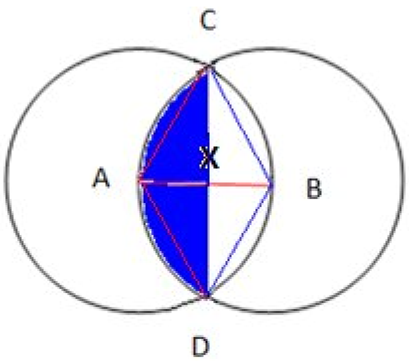
15. Congruent circles of centers A and B intersect such that AB is a radius of each circle. If AB= 4 cm, what is the number of square centimeters in the area ACBD that is common to the two circles?



(A) $\frac{16\pi}{3} - 4\sqrt{3}$ (C) $\frac{16\pi}{3} - 8\sqrt{3}$

(B) $\frac{32\pi}{3} - 4\sqrt{3}$ (D) $\frac{32\pi}{3} - 8\sqrt{3}$

Solution:



If $AB = 4$ then $BC = 4$ because radii of the same circle are equal

$BX = \frac{4}{2} = 2$

$CD \perp AB$ Why? Because CD and AB are diagonals of rhombus $ACBD$. Diagonals of a rhombus are perpendicular and bisect each other

$m\angle CXB = 90^{O}$

$\angle BCX = 30^{O}$ because BC is twice BX

So $\angle CBX = 60^{O}$

Therefore $\angle CBD = 120^{O}$ by symmetry

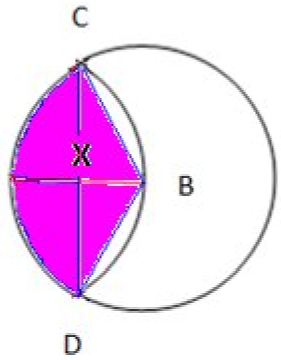
Recall:

In a 30^{O}-60^{O}-90^{O} triangle.

The hypotenuse is twice the side opposite of the 30^{O} angle.

Area of shaded part = area of the sector with central angle $\angle CBD$ minus area of ΔCBD

Area of sector:



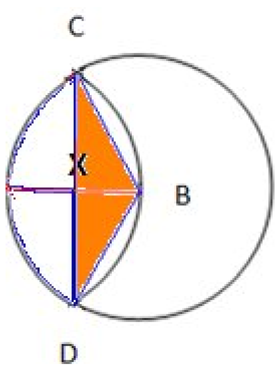
Area of sector with central angle $\angle CBD$

$= \pi r^2 \left(\dfrac{\theta}{360^O} \right)$

$= \pi (4^2) \left(\dfrac{120^O}{360^O} \right)$

$= \dfrac{16\pi}{3}$

Area of ΔCBD:



If $BX = 2$ then $CX = 2\sqrt{3}$

Therefore $CD = 4\sqrt{3}$

Area of ΔCBD = $\dfrac{\text{base} \times \text{height}}{2}$

$= \dfrac{CD \bullet BX}{2}$

$= \dfrac{4\sqrt{3} \bullet 2}{2}$

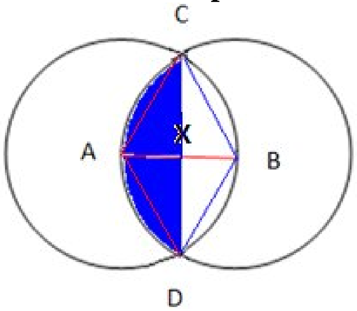
$= 4\sqrt{3}$

Recall:

In a 30^O-60^O-90^O triangle.

The side opposite of the 60^O angle is $\sqrt{3}$ times the length of the side opposite of the 30^O angle.

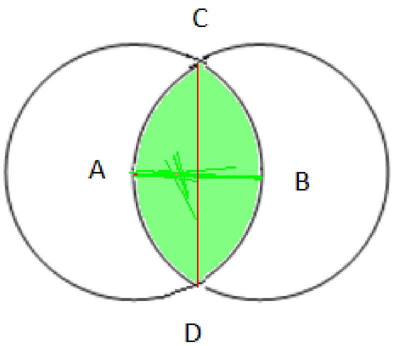
Area of shaded part:



Area of shaded part = area of the sector with central angle $\angle CBD$ minus area of ΔCBD

$$= \frac{16\pi}{3} - 4\sqrt{3}$$

Area ACBD that is common to the two circles:



area ACBD that is common to the two circles

$$= 2\times\left(\frac{16\pi}{3} - 4\sqrt{3}\right)$$

$= \frac{32\pi}{3} - 8\sqrt{3}$ because of symmetry

Answer: D

16. The value of $\left[2-3(2-3)^{-1}\right]^{-1}$ is

(A) 5 (C) 1/5

(B) -5 (D) -1/5

Solution:

$$\left[2-3(2-3)^{-1}\right]^{-1}$$

$$=\left[2-3(-1)^{-1}\right]^{-1}$$

$$=[2-3(-1)]^{-1}$$

$$=[2+3]^{-1}$$

$$=5^{-1}$$

$$=\frac{1}{5}$$

Answer: C

17. What is the sum of the first 100 positive odd integers?

(A) 25000 (C) 20000

(B) 50000 (D) 10000

Solution:

$S_n = \frac{n}{2}(a_1 + a_n)$ [This is the formula for the first n terms of an arithmetic series]

The 100th positive odd integer is 2(100) – 1 = 199

So $S_{100} = \frac{100}{2}(1+199)$

$= 50(200)$

$= 10000$

Answer: D

18. Sa orasan ni Ginoong Abad ay 5:14 a.m. nang siya ay umalis ng bahay. Bumalik siya makalipas ang 7 oras at 11 minuto, subalit nagtaka siya dahil 4:33 pa lang ng umaga sa orasan. Naalala niyang nagkaroon pala ng rotating brown out sa kanilang lugar kaya nang bumalik ang kuryente ay kusang bumalik sa oras na 12:00 a.m. ang orasan. Anong oras ng umaga bumalik ang kuryente?

(A) 8:52 (C) 5:52
(B) 6:52 (D) 7:52

Solution:

The time 7 hours and 11 minutes after 5:14 a.m. is 12:25p.m.

The time 4 hours and 33 minutes before 12:25 p.m. is 7:52 a.m.

Answer: D

19. Express in lowest terms: $\frac{n^3+7n^2}{n^2-2n-63}$

(A) $\frac{n}{n-9}$ (B) $\frac{n+7}{n-9}$ C) $\frac{n^2}{n-9}$ (D) $\frac{n^2}{n+7}$

Solution:

Factor the numerator and denominator and cancel the common factors

$$\frac{n^3+7n^2}{n^2-2n-63} = \frac{n^2\cancel{(n+7)}}{\cancel{(n+7)}(n-9)}$$

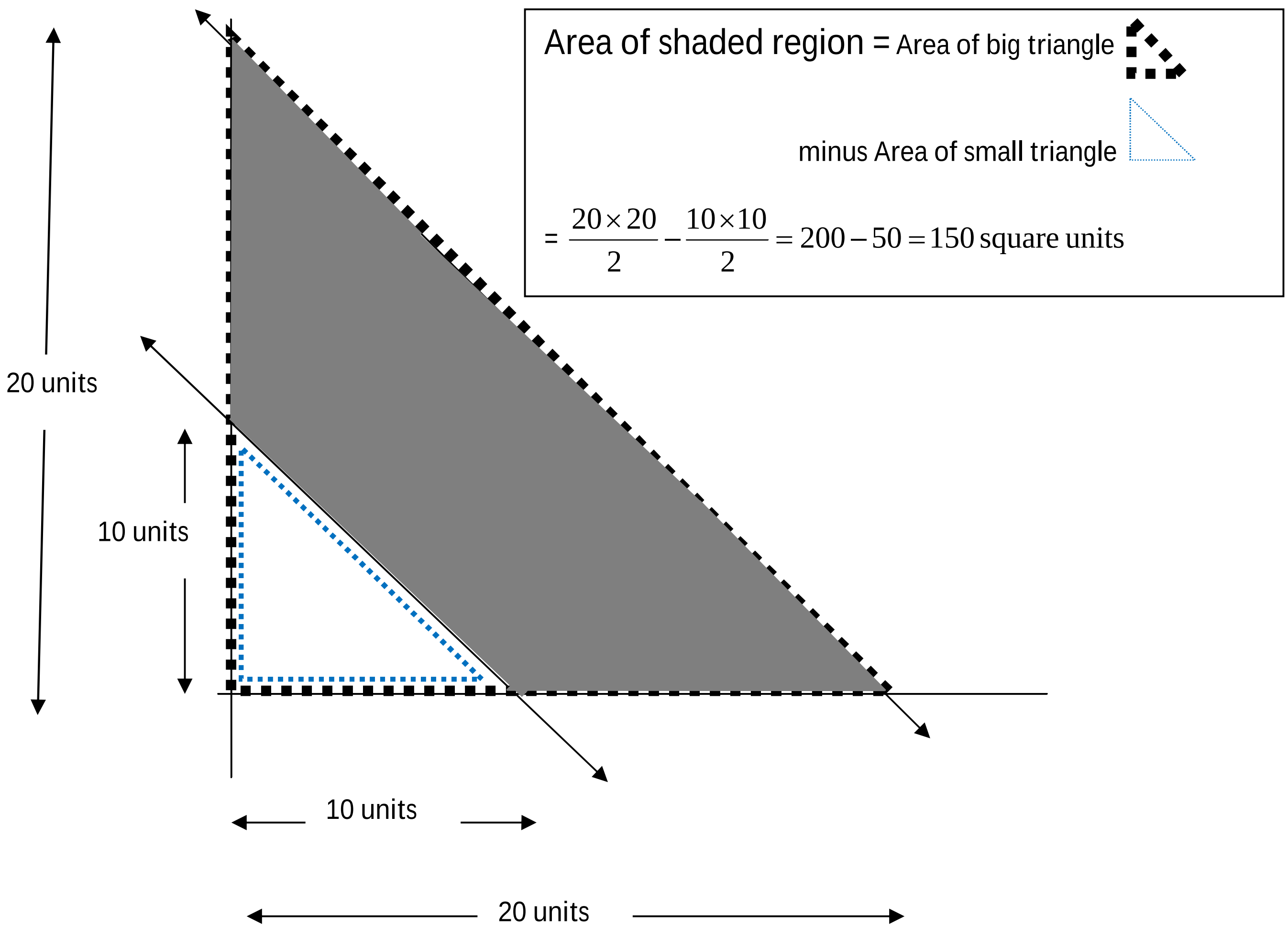
$$= \frac{n^2}{n-9}$$

Answer: C

20. How many square units are in the area defined by the set of points (x,y) in the first quadrant which satisfies $10 \le x + y \le 20$?

(A) 150 B) 152 (C) 154 (D) 144

Solution:



Answer: A

21. Fibonacci gave a famous problem about reproducing rabbits. After n months, there is a population of r_n rabbits, where $r_0 = 1$, $r_1 = 2$, $r_2 = 3$ and so on. What was the rabbit population at n=12 months?

(A) 55 (B) 89 (C) 144 (D) 233

Solution:

The 12th term of the sequence 1,2,3,5,8,13,21,34,55,89,144,233 is 233

Answer: D

22. For what value of x does $1+2+3+4+5+\ldots+x=171$?

(A) 17 (B) 16 (C) 18 (D) 21

Solution:

$$\frac{x(x+1)}{2}=171$$

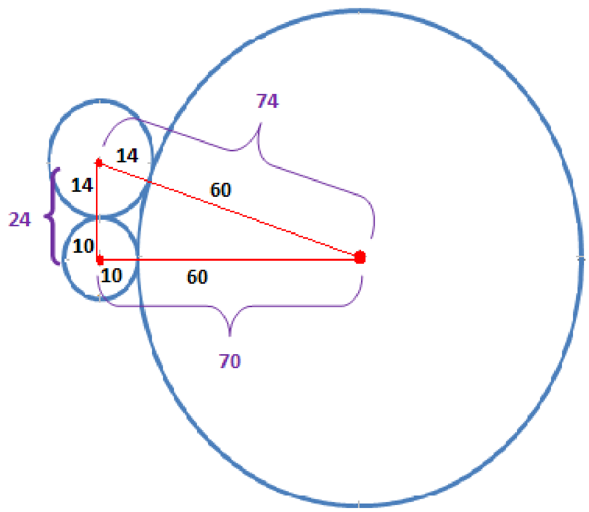
$\Rightarrow x=18$, we disregard the negative root

Answer: C

23. Three circles of radii 10, 14 and 60 are tangent to each other such that the center of each circle is outside the two other circles. Find the number of square units in the area of the triangle whose vertices are the centers of the three circles?

(A) 820 (B) 840 (C) 830 (D) 850

Solution:



The sides of a triangles are 24, 70, and 74 units. To get the area of the triangle we can use the Heron's formula.

$A=\sqrt{s(s-a)(s-b)(s-c)}$, where a,b, and c are the lengths of the triangle and s is the semi-perimeter of the triangle so $s=\frac{a+b+c}{2}$

In the triangle above $s=\frac{24+70+74}{2}=84$

$A=\sqrt{84(84-24)(84-70)(84-74)}$

$=\sqrt{84(60)(14)(10)}$

$=840$

$\sqrt{84(60)(14)(10)}$

$=\sqrt{84(10)(60)(14)}$

$=\sqrt{840(840)}$

$=\sqrt{840^2}$

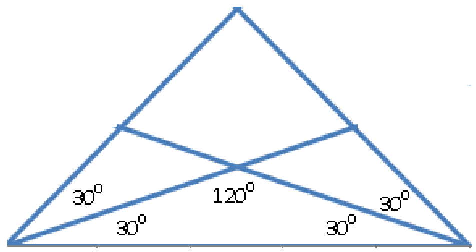
$=840$

Answer: B

24. What is the measure of the obtuse angle formed by two intersecting angle bisectors of an equilateral triangle?

(A) 150° (B) 110° (C) 120° (D) 100°

Solution:



Answer: C

25. We select 6 numbers at random, with replacement, from the set of integers from 1 to 300 inclusive. What is the probability that the product of the 6 numbers is even?

(A) $^{31}/_{32}$ (B) 1 (C) $^{61}/_{64}$ (D) $^{63}/_{64}$

Solution:

Let A= event that the product of 6 numbers selected at random , with replacement, from the set of integers from 1 to 300 inclusive is even.

So A^C = event that the product of 6 numbers selected at random, with replacement, from the set of integers from 1 to 300 inclusive is odd.

The product of 6 numbers is odd if each number is an odd number.

$$P(A^C) = \left(\frac{150}{300}\right)^6$$

$$= \left(\frac{1}{2}\right)^6$$

$$= \frac{1}{64}$$

Therefore

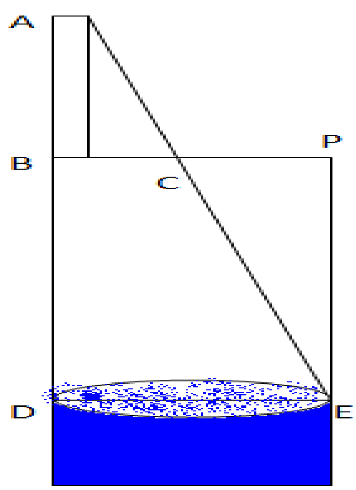
$$P(A) = 1 - P(A^C)$$
$$= 1 - \frac{1}{64}$$
$$= \frac{63}{64}$$

Answer: D

26. A deep well 5 feet in diameter is of unknown depth (to the water level). If a 5-foot post is erected at the edge of the well, the line of sight from the top of the post to the edge of the water surface below will pass through a point 0.4 feet from the lip of the well below the post. What is the depth of the well (to the surface of the water)?

(A) 37.5 ft (B) 47.5 ft (C) 57.5 ft (D) 67.5 ft

Solution:



The figure is not drawn into scale.

$AB = 5\ ft$

$BP = 5\ ft$

$BC = 0.4\ ft$

Let $BD = x\ ft$

$\Delta ABC \sim \Delta ADE$

Ratio of corresponding sides of similar triangles are equal

$$\frac{AB}{BC} = \frac{AD}{DE}$$

$$\frac{5}{0.4} = \frac{5+x}{5}$$

$$0.4x + 2 = 25$$

$$x = 57.5$$

Answer: C

27. Find integers p and q such that $\frac{4+3\sqrt{3}}{2+\sqrt{3}}$ is a root (zero) of the quadratic polynomial x^2+px+q.

(A) p = -2 and q =11

(B) p = -6 and q = 8

(C) p = -12 and q = 5

(D) p = -4 and q=9

Solution:

First we simplify $\frac{4+3\sqrt{3}}{2+\sqrt{3}}$

$$\frac{4+3\sqrt{3}}{2+\sqrt{3}} \bullet \frac{2-\sqrt{3}}{2-\sqrt{3}} = \frac{8-4\sqrt{3}+6\sqrt{3}-3(3)}{4-3}$$

$$= \frac{-1+2\sqrt{3}}{1}$$

$$= -1+2\sqrt{3}$$

The other root of the quadratic polynomial is the conjugate of $-1+2\sqrt{3}$ which is $-1-2\sqrt{3}$.

Sum of roots = $(-1+2\sqrt{3})+(-1-2\sqrt{3})=-2$

Product of roots = $(-1+2\sqrt{3})(-1-2\sqrt{3})=(-1)^2-(2\sqrt{3})^2 = 1-12=-11$

Therefore the quadratic equation is

$$x^2 - (\text{sum of roots})\,x + (\text{product of roots}) = 0$$

$$x^2+2x-11=0$$

$p=2$ and $q=-11$

Bonus: No answer in the given choices

28. Barb has two watches, one of which loses 6 seconds every 24 hours and the other gains 1 second per hour. He sets both of them to the correct time at 8:00 p.m. How many hours will pass before the positive difference in the time shown is 4 hours?

(A) 11484 (B) 11520 (C) 11511 (D) 11518

Solution:

After 1 hour the difference is $1-\left(-\frac{6}{24}\right)=\frac{5}{4}$ seconds

$$\left(\frac{4\text{ hours}}{5/4\text{ seconds}}\right)\text{hours}=\left(\frac{16}{5}\ \frac{\cancel{\text{hours}}}{\cancel{\text{seconds}}}\times\frac{60\ \cancel{\text{seconds}}}{1\ \cancel{\text{minute}}}\times\frac{60\ \cancel{\text{minutes}}}{1\ \cancel{\text{hour}}}\right)\text{hours}=11520\text{ hours}$$

Answer: B

29. Kung ang $3^x=20$, ano ang halaga ng 9^x?

(A) 30 (B) 400 (C) 180 (D) 729

Solution:

$3^x=20$, we get the square of both sides

$$3^{2x}=20^2$$

$$\Rightarrow 9^x=400$$

Answer: B

30. Kayang tapusin ng unang pintor ang trabaho sa loob ng 7 oras. Samantalang ang ikalawang pintor ay kayang tapusin ang parehong trabaho sa loob ng 13 oras. Ilang minuto matatapos ang trabaho kung magtutulungan ang dalawang pintor?

(A) 273 (B) 276 (C) 270 (D) 275

Solution:

$\frac{1}{7}x+\frac{1}{13}x=1$ multiply both sides by LCD = 91

$13x+7x=91$

$20x=91$

$x=\frac{91}{20}$ hours

$=273$ minutes

Answer: A

31. Summa Cum Laundry Inc. now has 5000 customers. Its number of customers increases by 20 % each year. At the end of how many full years will its number of customers first exceed 10,000?

(A) 4 (B) 3 (C) 5 (D) 6

Solution:

Number of customers at the end of	
1^{st} year	$5000\times1.2=6000$
2^{nd} year	$6000\times1.2=7200$
3^{rd} year	$7200\times1.2=8640$
4^{th} year	$8640\times1.2=10368>10000$

Answer: A

32. Which of the following is equal to $\sqrt[3]{54}+\sqrt[6]{4}$.

(A) $\sqrt[3]{128}$ (B) $\sqrt[3]{56}$ (C) $\sqrt[9]{216}$ (D) $\sqrt[6]{112}$

Solution:

First term:

$$\sqrt[3]{54}=\sqrt[3]{27\cdot 2}$$
$$=\sqrt[3]{27}\cdot\sqrt[3]{2}$$
$$=3\sqrt[3]{2}$$

Second term:

$$\sqrt[6]{4}=\sqrt[6]{2^2}$$
$$=2^{\frac{2}{6}}$$
$$=2^{\frac{1}{3}}$$
$$=\sqrt[3]{2}$$

Therefore $\sqrt[3]{54}+\sqrt[6]{4}=3\sqrt[3]{2}+\sqrt[3]{2}=4\sqrt[3]{2}=\sqrt[3]{64}\cdot\sqrt[3]{2}=\sqrt[3]{64\cdot 2}=\sqrt[3]{128}$

Answer: A

33. On a number line, how many integers are no more than 8 units from 20 and also at least 10 units from 35?

(A) 18 (B) 12 (C) 15 (D) 14

Solution:

A= Set of integers no more than 8 units from 20 = $\{12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27,28\}$

B = Set of integers at least 10 units from 35 = $\{...,9,10,11,12,13,14,15,16,17,18,19,20,21,22,23,24,25\}\bigcup\{45,46,47,48,49,50,...\}$

$$A\bigcap B=\{12,13,14,15,16,17,18,19,20,21,22,23,24,25\}$$

The number of elements in $A\bigcap B$ is 14.

Answer: D

34. Ang limang magkakasunod na integer ay may average na x . Ano ang pinakamababang integer?

(A) x-2 (B) x-3 (C) x-4 (D) x-5

Solution:

Since there are 5 consecutive integers and the average is x then there are 2 integers above x and 2 integers below x. The integers are $x-2, x-1, x, x+1, x+2$. The smallest integer is $x-2$.

Answer: A

35. The lines $y=(k-5)x+5$ and $y=-2x+7$ are perpendicular if k=

(A) 11/2 (B) 5 (C) -2/9 (D) 9/2

Solution:

The product of the slopes should be equal to -1 because they are perpendicular.

$$(-2)(k-5)=-1$$
$$-2k+10=-1$$
$$-2k=-11$$
$$k=\frac{11}{2}$$

Answer: A

36. Find the length of the radius of the circle $x^2+4x+y^2-6y-3=0$.

(A) 2 (B) 3 (C) 4 (D) 9

Solution:

$$x^2+4x+y^2-6y-3=0$$
$$(x^2+4x)+(y^2-6y)=3$$
$$(x^2+4x+4)+(y^2-6y+9)=3+4+9$$
$$(x+2)^2+(y-3)^2=16$$

$$r=4$$

Recall:

the equation of the circle is

$(x-h)^2+(y-k)^2=r^2$

center: (h,k)

length of radius: r

Answer: C

37. Given that $\begin{cases} x+y+z=0 \\ x^2+y^2+z^2=1 \end{cases}$, find the value of $x^4+y^4+z^4$.

(A) $\frac{2}{5}$ (B) $\frac{1}{3}$ (C) 3 (D) $\frac{1}{2}$

Solution:

First equation:

$x+y+z=0$ get the square of both sides

$$\underbrace{x^2+y^2+z^2}_{1}+2xy+2xz+2yz=0$$

$2xy+2xz+2yz=-1$ divide both sides by 2

$xy+xz+yz=\frac{-1}{2}$ get the square of both sides

$$x^2y^2+x^2z^2+y^2z^2+\underbrace{2x^2yz+2xy^2z+2xyz^2}=\frac{1}{4}$$

$$2xyz(x+y+z)=2(0)=0$$

So $x^2y^2+x^2z^2+y^2z^2=\frac{1}{4}$

Second equation:

$x^2+y^2+z^2=1$ get the square of both sides

$$x^4+y^4+z^4+2x^2y^2+2x^2z^2+2y^2z^2=1$$

$$x^4+y^4+z^4+2\underbrace{\left(x^2y^2+x^2z^2+y^2z^2\right)}_{1/4}=1$$

Therefore $x^4+y^4+z^4=\frac{1}{2}$

Answer: D

38. Simplify: $\dfrac{\frac{1}{b}+\frac{1}{c}}{b^2-c^2}$

(A) $\dfrac{1}{bc(b-c)}$

(B) $\dfrac{bc}{b-c}$

(C) $\dfrac{1}{bc}$

(D) $\dfrac{b-c}{bc}$

Solution:

Multiply the numerator and denominator by bc the result is,

$$\frac{\frac{1}{b}+\frac{1}{c}}{b^2-c^2}\bullet\frac{bc}{bc}=\frac{c+b}{bc\left(b^2-c^2\right)}$$, factor the denominator and cancel the common factor

$$=\frac{\cancel{b+c}}{bc(b-c)\cancel{(b+c)}}$$
$$=\frac{1}{bc(b-c)}$$

Answer: B

39. Let $f(x)=\begin{cases} x-11 & \text{if } x\le 6 \\ x+11 & \text{if } x>6 \end{cases}$. Find $f(12)+f(6)$

(A) -4 (B) 28 (C) 40 (D) 18

Solution:

$f(12)=12+11=23$

$f(6)=6-11=-5$

$f(12)+f(6)=23+(-5)=18$

Answer: D

40. The graph of $16x-2y=48$ intersects the y-axis at the point (a,b).What is the sum of a and b?

(A) -4 (B) -24 (C) 32 (D) 12

Solution:

To get the y-intercept of $16x-2y=48$ we let $x=0$ and solve for y.

$16(0)-2y=48$

So the y-intercept is $(0,-24)$

Therefore $a+b=0+(-24)=-24$

Answer: B

41. For what values of the variable x does the following inequality hold?

$$\frac{4x^2}{\left(1-\sqrt{1+2x}\right)^2}<2x+9?$$

(A) $x\le -\frac{1}{2}$ or $x>\frac{45}{8}$

(B) $x\ge -\frac{1}{2}$ $except\ x=0$

(C) $-\frac{1}{2}\le x<\frac{45}{8}$ $except\ x=0$

(D) $x<\frac{45}{8}$ $except\ x=0$

Solution:

The radicand (expression inside the square root sign) $1+2x$ should be non negative

$1+2x\ge 0$

$x\ge \frac{-1}{2}$ *

The denominator $\left(1-\sqrt{1+2x}\right)^2$ should not be equal to zero.

We solve $\left(1-\sqrt{1+2x}\right)^2=0$

$$1-\sqrt{1+2x}=0$$

$$\sqrt{1+2x}=1$$

$$1+2x=1$$

$$2x=0$$

$$x=0$$

So if x=0, then the expression on the left side of the inequality is undefined

Therefore $x\neq 0$ **

Multiply both sides of the inequality $\dfrac{4x^2}{\left(1-\sqrt{1+2x}\right)^2}<2x+9$ by $\left(1-\sqrt{1+2x}\right)^2$, the result is

$$4x^2<(2x+9)\left(1-\sqrt{1+2x}\right)^2$$ We don't change the direction of the inequality because we multiply both sides by a positive number

$$4x^2<(2x+9)\left(2x+2-2\sqrt{1+2x}\right)$$

$$\cancel{4x^2}<\cancel{4x^2}+22x+18-(4x+18)\sqrt{1+2x}$$

$$(4x+18)\sqrt{1+2x}<22x+18$$

Let $y=\sqrt{1+2x}$

So $y^2=1+2x$

Also $4x+18=2y^2+16$

And $22x+18=11y^2+7$

The inequality $(4x+18)\sqrt{1+2x}<22x+18$ in terms of y is

$(2y^2+16)(y)<11y^2+7$

$2y^3-11y^2+16y-7<0$, we can use synthetic division to find the factors of the left side

$(y-1)^2(2y-7)<0$ divide both sides by $(y-1)^2$ a positive number so don't change the direction of the inequality [So $y\neq 1$]

$2y-7<0$

$y<\frac{7}{2}$ express this inequality in terms of the variable x

$\sqrt{1+2x}<\frac{7}{2}$ square both sides [the resulting inequality is true because both sides are positive]

$1+2x<\frac{49}{4}$

$2x<\frac{45}{4}$

$x<\frac{45}{8}$ ***

$x\geq\frac{-1}{2}$ *

$x\neq 0$ **

$x<\frac{45}{8}$ ***

The intersection of the above inequalities is

$$-\frac{1}{2} \leq x < \frac{45}{8} \quad except\ x = 0$$

Answer: C

42. All the sides of a given triangle are whole numbers. The perimeter of the triangle is 110 inches and one side has measure 29 inches. What is the fewest number of inches that can be the length of one of the remaining sides?

(A) 30 (C) 26

(B) 24 (D) 27

Solution:
Let a and b be the length of the other two sides where $a > b$

The perimeter is 110 so $a + b + 29 = 110$ or $a + b = 81$

$$a - b < 29$$

$$(81 - b) - b < 29$$

$$-2b < -52$$

$$b > 26$$

The smallest possible length of one of the remaining sides is 27 inches.
Answer: D

43. If a regular polygon has 27 diagonals, how many sides does it have?

(A) 12 (C) 9

(B) 6 (D) 7

Solution:
$$\frac{n(n-3)}{2} = 27$$

$$\Rightarrow n(n-3) = 54$$

$\Rightarrow n^2 - 3n - 54 = 0$

$\Rightarrow (n-9)(n+6) = 0$

$\Rightarrow n = 9$ or $n = -6$
disregard $n = -6$

Answer: C

44. If a and b are each chosen from the set $\{1,2,3,5,10\}$, what is the largest possible value of $\frac{a}{b} + \frac{b}{a}$?

(A) 20

(B) $12\frac{1}{2}$

(C) $10\frac{1}{10}$

(D) $2\frac{1}{2}$

Solution:

	b=1	b=2	b=3	b=5	b=10
a=1		2 1/2	3 1/3	5 1/5	10 1/10
a=2	2 1/2		2 1/6	2 9/10	5 1/5
a=3	3 1/3	2 1/6		2 4/15	3 19/30
a=5	5 1/5	2 9/10	2 4/15		2 1/2
a=10	10 1/10	5 1/5	3 19/30	2 1/2	

the largest possible value of $\frac{a}{b} + \frac{b}{a}$ is $10\frac{1}{10}$.

Answer: C

45. Which of the following relations are functions?

I. $x^2 + y^2 = 4$

II. $x - y = 2x$

III. $y = \sqrt{4-x}$

IV. $x = y^2 - 1$

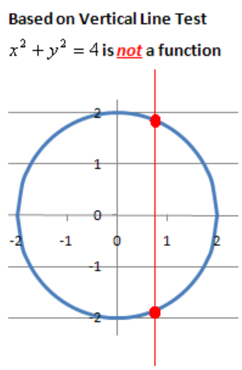
V. $y = x^2 - 1$

VI. $x^2 = y^2$

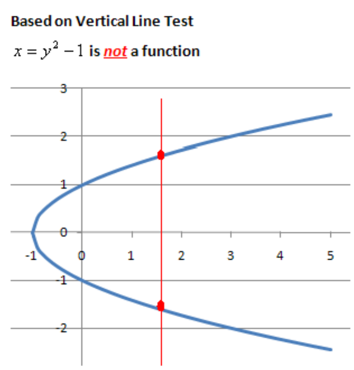
(A) I, II, III
(B) II, III, IV
(C) II, III, V
(D) I, IV, VI

Solution:

I. $x^2 + y^2 = 4$

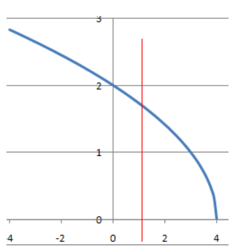


IV. $x = y^2 - 1$



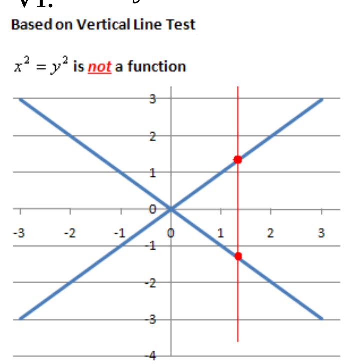
II. $x - y = 2x$ is a linear ***function***

III. $y = \sqrt{4-x}$ is a ***function***



V. $y = x^2 - 1$ is a quadratic ***function***

VI. $x^2 = y^2$



Answer: C

46. Suppose $x+y=3$, $x^2+y^2=13$. Find the value of xy.

(A) 39 (B) -6 (C) 24 (D) -2

Solution:

$(x+y)^2 = x^2+2xy+y^2$

$\Rightarrow xy = \dfrac{(x+y)^2-(x^2+y^2)}{2}$

$\Rightarrow xy = \dfrac{(3)^2-(13)}{2}$

$\Rightarrow xy = -2$

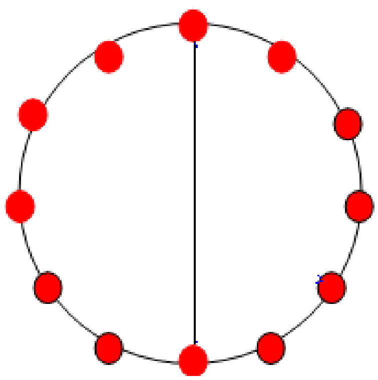
Answer: D

47. Suppose there are 12 points on a circle, equally spaced. Of all triangles having all their vertices at three of these 12 points, how many are of the type $30^\circ - 60^\circ - 90^\circ$?

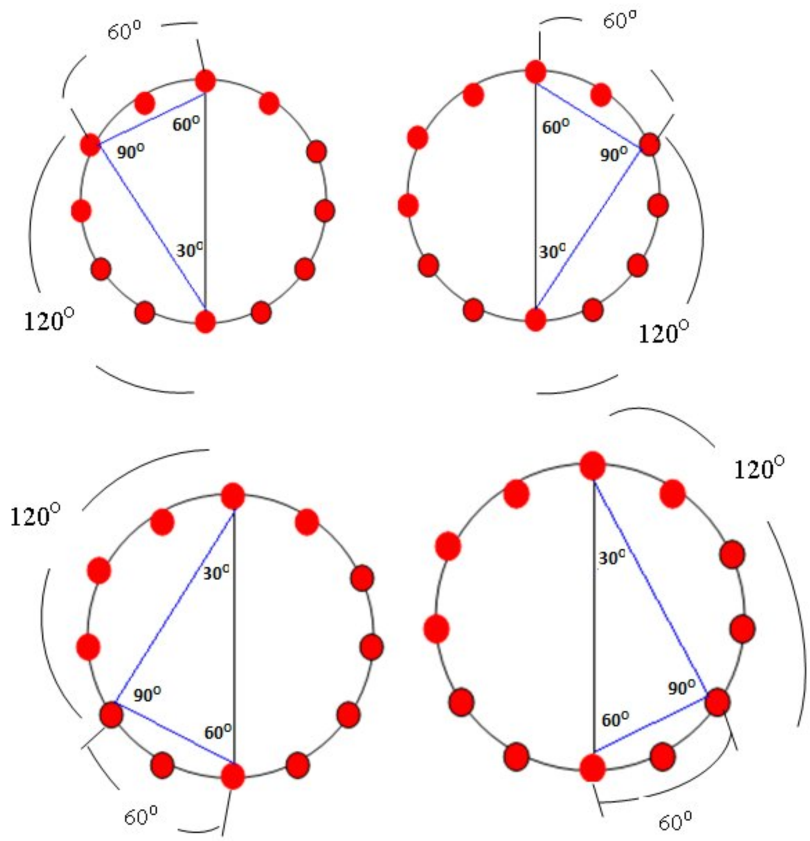
(A) 24 (B) 36 (C) 72 (D) none of these

Solution:

If a right angled triangle is inscribed in a circle then the hypotenuse is also the diameter of the circle.



Using the diameter above the number of $30^O - 60^O - 90^O$ triangles that can be formed is four.



There are 6 diameters that can be formed using 12 points that are spaced evenly around a circle. Therefore the number of triangles having all their vertices at three of these 12 points, that are of the type $30^\circ - 60^\circ - 90^\circ$ is $6 \times 4 = 24$.

Answer: A

48. Find all the values of x satisfying the given conditions: $y_1 = \dfrac{x-3}{3}$, $y_2 = \dfrac{x-3}{4}$, and $y_1 - y_2 = 3$

(A) 39 (B) 49 (C) 59 (D) 69

Solution:

$y_1 - y_2 = 3$

$\frac{x-3}{3}-\frac{x-3}{4}=3$ multiply both sides by 12

$4(x-3)-3(x-3)=36$

$4x-12-3x+9=36$

$x-3=36$

$x=39$

Answer: A

49. The lines $y=2x$ and $2y=-x$ are

(A) parallel (B) perpendicular (C) horizontal (D) vertical

Solution:

The slopes of the lines $y=2x$ and $2y=-x$ are 2 and $-\frac{1}{2}$ respectively.
The slopes are negative reciprocal of each other. Therefore the two lines are perpendicular.

Answer: B

50. Suppose the following two quadratic equations

$x^2-5x+k=0$ and $x^2-9x+3k=0$

have a non-zero root in common. What is the value of k?

(A) 4 (B) 5 (C) 6 (D) 7

Solution:

Recall:

In the quadratic equation

$ax^2 + bx + c = 0$, where $a \neq 0$

Sum of roots = $-\dfrac{b}{a}$

Product of roots = $\dfrac{c}{a}$

Let p and q be the roots of $x^2 - 5x + k = 0$

q and r be the roots of $x^2 - 9x + 3k = 0$

q is the common root

$p + q = 5$ $\qquad\qquad q + r = 9$

$pq = k$ * $\qquad\qquad qr = 3k$ **

Using * and ** $\dfrac{\cancel{q}r}{p\cancel{q}} = \dfrac{3\cancel{k}}{\cancel{k}}$

$r = 3p$

So the equation $q + r = 9$ becomes $q + 3p = 9$

$\Rightarrow q = 9 - 3p$, substitute this in the equation $p + q = 5$

It becomes

$p + (9 - 3p) = 5$

$-2p = -4$

$p = 2$, therefore $q = 3$

$k = pq = (2)(3) = 6$

Answer: C

(A) 4 (B) 5 (C) 6 (D) 7

51. Let x, y, and z represent lengths of the sides of a right triangle with y<z<x. If $L = 6x^2 + 15y^2 + 15z^2$, find the value of $\dfrac{L}{y^2 + z^2}$.

(A) 12 (B) 15 (C) 18 (D) 21

Solution:

The length of the biggest side in a right triangle is x. Using Phytagorean theorem $y^2 + z^2 = x^2$.

$$\frac{L}{y^2 + z^2} = \frac{6x^2 + 15y^2 + 15z^2}{y^2 + z^2}$$

$$=\frac{6x^2+15(y^2+z^2)}{y^2+z^2}$$

$$=\frac{6x^2+15x^2}{x^2}$$

$$=\frac{6x^2+15x^2}{x^2}$$

$$=\frac{21\cancel{x^2}}{\cancel{x^2}}$$

$$=21$$

Answer: D

52. The arithmetic mean of 14 numbers is what percent of the sum of the same 14 numbers?. Express your answer as a decimal to the nearest hundredth.

(A) 7.69% (B) 7.14% (C) 8.33% (D) 5.88%

Solution:

Let x be the sum of the 14 numbers

Arithmetic mean = $\frac{x}{14}$

$$\frac{x/14}{x}=\frac{1}{14}$$

$$\approx 7.14\%$$

Answer: B

53. Let $f(x)=x/\ln x$ and $\pi(x)$ denotes the number of primes less than or equal to x . When Gauss was 15 years old, he conjectured that as x goes to infinity, the ratio of $\pi(x)$ to $f(x)$ approaches one. Use Gauss's estimate to estimate the percentage of positive integers smaller than 10^{100} which are prime. (Remark: the natural logarithm of 10 is about 2.3 .)

(A) 10% (B) 3% (C) 0.4% (D) 0.05%

Solution:

percentage of positive integers smaller than 10^{100} which are prime

$$= \frac{\text{number of positive integers smaller than } 10^{100} \text{ which are prime}}{10^{100}}$$

$$\approx \frac{f\left(10^{100}\right)}{10^{100}}$$ because as x goes to infinity $\pi(x) \approx f(x)$

$$= \frac{10^{100} / \ln\left(10^{100}\right)}{10^{100}}$$

$$= \frac{1}{\ln\left(10^{100}\right)}$$

$$= \frac{1}{100 \ln 10}$$

$$\approx \frac{1}{100(2.3)}$$

$$= \frac{1}{230}$$

$\approx 0.4\%$

Answer: C

54. Given that $81^m = 3$ and $m^n = 64$. Find the value of mn.

(A) -1/2 (C) -3/4

(B) -1 (D) -1/4

Solution:

$$81^m = 3$$
$$\Rightarrow 3^{4m} = 3^1$$
$$\Rightarrow 4m = 1$$
$$\Rightarrow m = \frac{1}{4}$$

$$m^n = 64$$
$$\Rightarrow \left(\frac{1}{4}\right)^n = 64$$
$$\Rightarrow 4^{-n} = 4^3$$
$$\Rightarrow n = -3$$

$$mn = -\frac{3}{4}$$

Answer: C

55. What is the sum of all the elements of this finite arithmetic sequence

3, 6, 9, 12, 15, 18, ... , 42 ?

(A) 314 (B) 313 (C) 318 (D) 315

Solution:

$$3+6+9+12+15+18+...+42 = 3(1+2+3+4+5+6+...+14)$$

We can use the formula for the sum of the first n natural numbers.

$$1+2+3+...+n = \frac{n\,(n+1)}{2}$$

$$3(1+2+3+4+5+6+...+14)=3\times\frac{14(15)}{2}$$

$$=315$$

Answer: D

56. The mean, median and mode for this set of data are all equal:
{ x, 68,30,58,52} . Find x .

(A) 52 (B) 68 (C) 58 (D) 30

Solution:

$$\frac{x+68+30+58+52}{5}=x$$

$$x+208=5x$$

$$4x=208$$

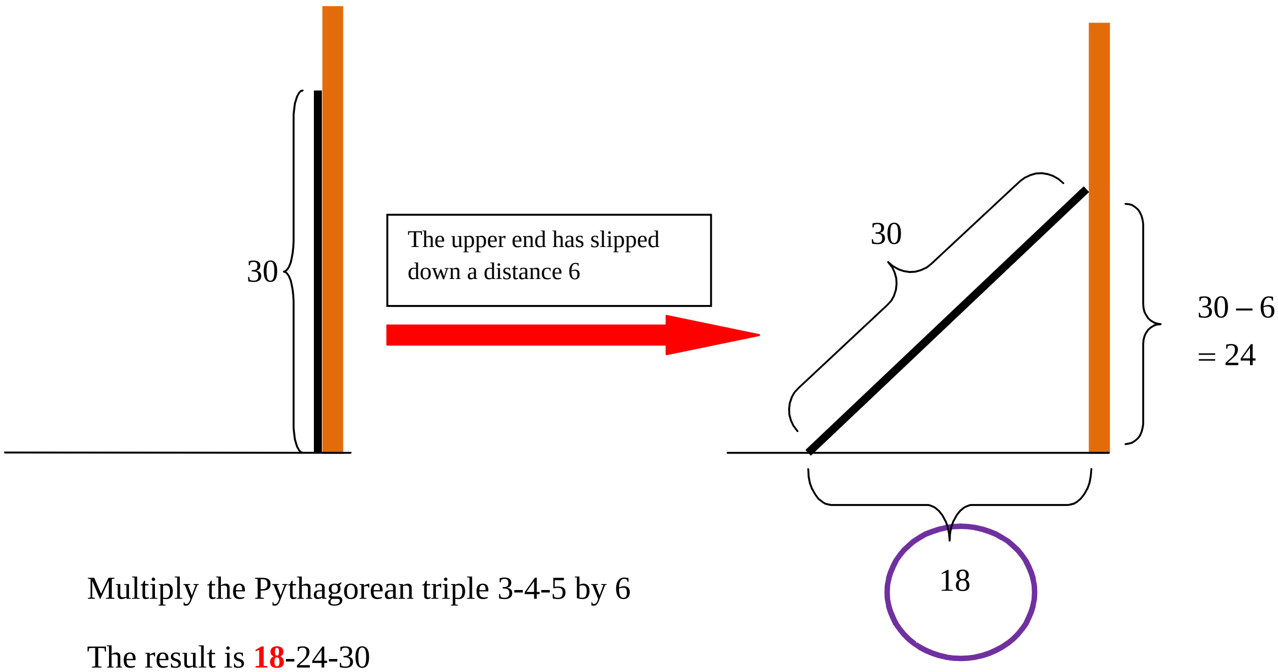
$$x=52$$

mean = median = mode = 52

Answer: A

57. In an old Babylonian tablet (circa 1700 B.C.) there is a problem: “A beam of length 30 stands against a wall. The upper end has slipped down a distance 6. How far did the lower end move?” The answer is

(A) 6 (B) 12 (C) 18 (D) 24



Multiply the Pythagorean triple 3-4-5 by 6

The result is **18**-24-30

Answer: C

58. Ang produkto ng dalawang buong numero ay 100,000. Kung ang alinman sa mga numero ay hindi multiple ng 10, ano ang kanilang suma(sum) ?

(A) 3158 (B) 3157 (C) 3159 (D) 3160

Solution:

$$100000 = 10^5$$

$$= (2 \times 5)^5$$

$$= 2^5 \times 5^5$$

$$= 32 \times 3125$$

$$32 + 3125 = 3157$$

Answer: B

59. Solve for x: $\left(\frac{19}{30}\right) + \left(\frac{11}{30}\right)^2 = \left(\frac{19}{30}\right)^2 + x$

(A) 11/30 (C) 2/5

(B) 1/3 (D) 3/10

Solution:

$$x = \frac{19}{30} + \left(\frac{11}{30}\right)^2 - \left(\frac{19}{30}\right)^2$$

$$x = \frac{19}{30} + \left(\frac{11}{30} - \frac{19}{30}\right)\left(\frac{11}{30} + \frac{19}{30}\right)$$

$$= \frac{19}{30} + \left(\frac{-8}{30}\right)(1)$$

$$= \frac{11}{30}$$

Recall:

$$a^2 - b^2 = (a-b)(a+b)$$

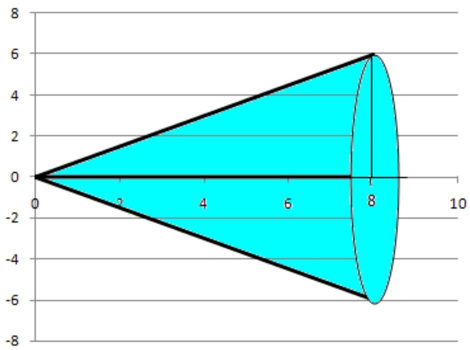
Answer: A

60. A triangle with vertices A(0,0), B(8,0), and C(8,6) is graphed in a coordinate plane. The triangular region determined by these points is rotated 360° about the x-axis forming a geometric solid. How many square units are in the total outside area of this geometric solid?

(A) 90π (C) 120π

(B) 84π (D) 96π

Solution:



Base area is:

$$A = \pi r^2$$

The surface area of a cone is therefore given by:

$$SA = \pi r^2 + \pi rs$$

$SA = \pi r(r+s)$ where r=radius of the base

h = height

s = length of the side

s can be calculated by using the Pythagorean theorem

$$s = \sqrt{r^2 + h^2}$$

r=6, h=8, $s = \sqrt{6^2 + 8^2} = 10$

$SA = 6\pi(6+10) = 96\pi$ **Answer: D**